

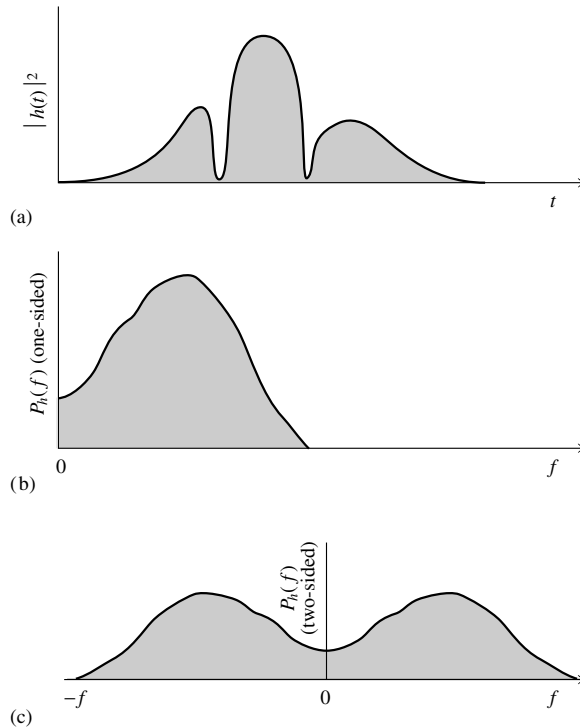


Figure 12.0.1. Normalizations of one- and two-sided power spectra. The area under the square of the function, (a), equals the area under its one-sided power spectrum at positive frequencies, (b), and also equals the area under its two-sided power spectrum at positive and negative frequencies, (c).

so that the total power is just the integral of $P_h(f)$ from $f = 0$ to $f = \infty$. When the function $h(t)$ is real, the two terms in (12.0.15) are equal, so $P_h(f) = 2|H(f)|^2$. Be warned that one occasionally sees PSDs defined without this factor two. These, strictly speaking, are called *two-sided power spectral densities*, but some books are not careful about stating whether one- or two-sided is to be assumed. We will always use the one-sided density given by equation (12.0.15). Figure 12.0.1 contrasts the two conventions.

If the function $h(t)$ goes endlessly from $-\infty < t < \infty$, then its total power and power spectral density will, in general, be infinite. Of interest then is the (*one- or two-sided*) *power spectral density per unit time*. This is computed by taking a long, but finite, stretch of the function $h(t)$, computing its PSD [that is, the PSD of a function that equals $h(t)$ in the finite stretch but is zero everywhere else], and then dividing the resulting PSD by the length of the stretch used. Parseval's theorem in this case states that the integral of the one-sided PSD-per-unit-time over positive frequency is equal to the mean square amplitude of the signal $h(t)$.

You might well worry about how the PSD-per-unit-time, which is a function of frequency f , converges as one evaluates it using longer and longer stretches of data. This interesting question is the content of the subject of “power spectrum estimation” and will be considered below in §13.4 – §13.7. A crude answer for now is, the PSD-per-unit-time converges to finite values at all frequencies *except* those where $h(t)$ has a discrete sine-wave (or cosine-wave) component of finite amplitude. At

those frequencies, it becomes a delta-function, i.e., a sharp spike, whose width gets narrower and narrower, but whose area converges to be the mean square amplitude of the discrete sine or cosine component at that frequency.

We have by now stated all of the analytical formalism that we will need in this chapter, with one exception: In computational work, especially with experimental data, we are almost never given a continuous function $h(t)$ to work with, but are given, rather, a list of measurements of $h(t_i)$ for a discrete set of t_i 's. The profound implications of this seemingly trivial fact are the subject of §12.1.

12.0.1 Higher-Order Statistics

The Wiener-Khinchin theorem, equation (12.0.13), along with the definition (12.0.11), shows that the power spectrum of a function is fully equivalent to the function's *two-point statistic*, that is, the expectation value of the product of the function at two different points separated by t . One can correspondingly define *higher-order statistics* in both the time and Fourier domains. For example, a function's *three-point correlation* is

$$\text{Corr3}(g, g, g) \equiv \int_{-\infty}^{\infty} g(\tau)g(\tau + t_1)g(\tau + t_2) d\tau \quad (12.0.16)$$

a function of the two variables t_1 and t_2 . The two-dimensional Fourier transform (§12.5) of equation (12.0.16) over t_1 and t_2 is called the *bispectrum*, a function of two frequencies f_1 and f_2 .

Higher-order statistics, including the bispectrum, can make visible non-Gaussian and nonlinear phenomena to which two-point statistics (and thus power spectra) are blind. However, they have the disadvantages of being often difficult to interpret and, because of the high powers of the signal that enter, highly susceptible to noise. On these grounds, we advise caution. Useful, if sometimes overly enthusiastic, references are [1,2,3].

CITED REFERENCES AND FURTHER READING:

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- Mendel, J.M. 1991, "Tutorial on Higher-Order Statistics (Spectra) in Signal Processing and System Theory: Theoretical Results and Some Applications," *Proceedings of the IEEE*, vol. 79, pp. 278–305.[2]
- Nikias, C.L., and Petropulu, A.P. 1993, *Higher-Order Spectra Analysis* (New Jersey: Prentice-Hall).[3]

12.1 Fourier Transform of Discretely Sampled Data

In the most common situations, function $h(t)$ is sampled (that is, its value is recorded) at evenly spaced intervals in time. Let Δ denote the time interval between consecutive samples, so that the sequence of sampled values is

$$h_n = h(n\Delta) \quad n = \dots, -3, -2, -1, 0, 1, 2, 3, \dots \quad (12.1.1)$$

The reciprocal of the time interval Δ is called the *sampling rate*; if Δ is measured in seconds, for example, then the sampling rate is the number of samples recorded per second.

12.1.1 Sampling Theorem and Aliasing

For any sampling interval Δ , there is also a special frequency f_c , called the *Nyquist critical frequency*, given by

$$f_c \equiv \frac{1}{2\Delta} \quad (12.1.2)$$

If a sine wave of the Nyquist critical frequency is sampled at its positive peak value, then the next sample will be at its negative trough value, the sample after that at the positive peak again, and so on. Expressed otherwise: *Critical sampling of a sine wave is two sample points per cycle*. One frequently chooses to measure time in units of the sampling interval Δ . In this case, the Nyquist critical frequency is just the constant $1/2$.

The Nyquist critical frequency is important for two related, but distinct, reasons. One is good news, and the other bad news. First the good news. It is the remarkable fact known as the *sampling theorem*: If a continuous function $h(t)$, sampled at an interval Δ , happens to be *bandwidth limited* to frequencies smaller in magnitude than f_c , i.e., if $H(f) = 0$ for all $|f| \geq f_c$, then the function $h(t)$ is *completely determined* by its samples h_n . In fact, $h(t)$ is given explicitly by the formula

$$h(t) = \Delta \sum_{n=-\infty}^{+\infty} h_n \frac{\sin[2\pi f_c(t - n\Delta)]}{\pi(t - n\Delta)} \quad (12.1.3)$$

This is a remarkable theorem for many reasons, among them that it shows that the “information content” of a bandwidth limited function is, in some sense, infinitely smaller than that of a general continuous function. Fairly often, one is dealing with a signal that is known on physical grounds to be bandwidth limited (or at least approximately bandwidth limited). For example, the signal may have passed through a physical component with a known, finite frequency response. In this case, the sampling theorem tells us that the entire information content of the signal can be recorded by sampling it at a rate Δ^{-1} equal to twice the maximum frequency passed by the amplifier (cf. equation 12.1.2).

Now the bad news. The bad news concerns the effect of sampling a continuous function that is *not* bandwidth limited to less than the Nyquist critical frequency. In that case, it turns out that all of the power spectral density that lies outside of

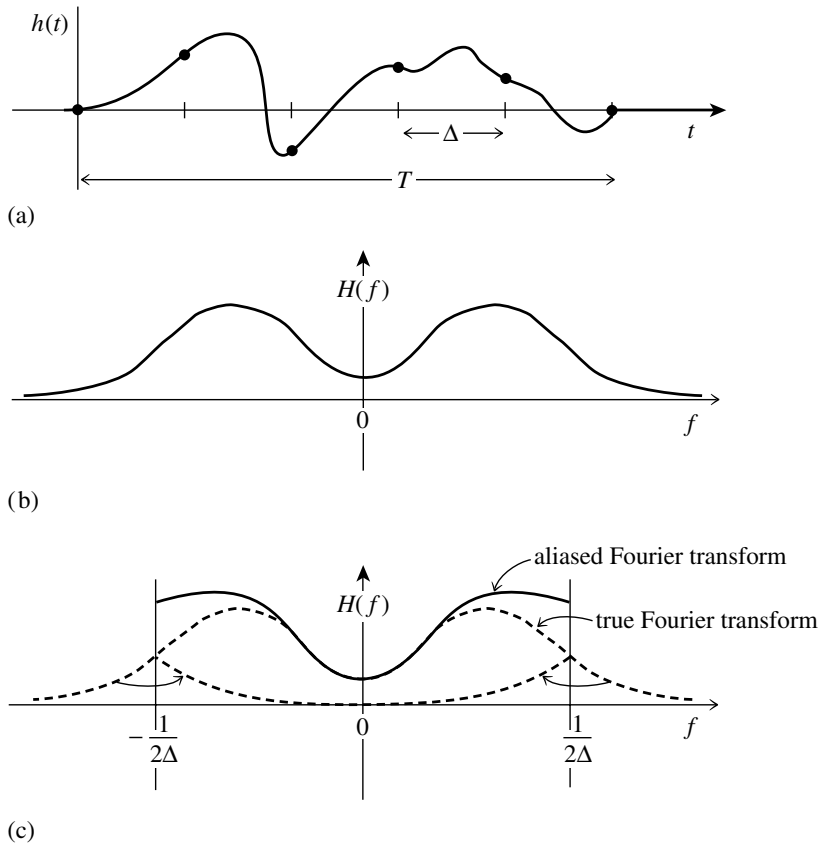


Figure 12.1.1. The continuous function shown in (a) is nonzero only for a finite interval of time T . It follows that its Fourier transform, whose modulus is shown schematically in (b), is not bandwidth limited but has finite amplitude for all frequencies. If the original function is sampled with a sampling interval Δ , as in (a), then the Fourier transform (c) is defined only between plus and minus the Nyquist critical frequency. Power outside that range is folded over or “aliased” into the range. The effect can be eliminated only by low-pass filtering the original function *before sampling*.

the frequency range $-f_c < f < f_c$ is spuriously moved into that range. This phenomenon is called *aliasing*. Any frequency component outside of the frequency range $(-f_c, f_c)$ is *aliased* (falsely translated) into that range by the very act of discrete sampling. You can readily convince yourself that two waves $\exp(2\pi i f_1 t)$ and $\exp(2\pi i f_2 t)$ give the same samples at an interval Δ if and only if f_1 and f_2 differ by a multiple of $1/\Delta$, which is just the width in frequency of the range $(-f_c, f_c)$. There is little that you can do to remove aliased power once you have discretely sampled a signal. The way to overcome aliasing is to (i) know the natural bandwidth limit of the signal — or else enforce a known limit by analog filtering of the continuous signal, and then (ii) sample at a rate sufficiently rapid to give at least two points per cycle of the highest frequency present. Figure 12.1.1 illustrates these considerations.

To put the best face on this, we can take the alternative point of view: If a continuous function has been competently sampled, then, when we come to estimate its Fourier transform from the discrete samples, we can *assume* (or rather we *might as well* assume) that its Fourier transform is equal to zero outside of the frequency range